Automated Fish Feeder

using ESP32

Mini Project – Product Development Lab

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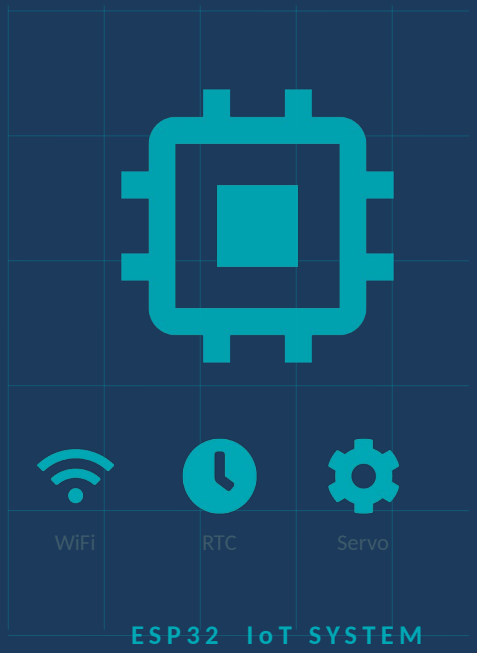

Introduction & Problem Statement

Why do we need this?

Fish are sensitive creatures that require regular, measured feeding to stay healthy. Missing a feeding — or overfeeding — can harm the fish and pollute the water.

The Problems

- Manual feeding is often skipped due to busy schedules or travel
- Irregular feeding causes overfeeding or underfeeding
- Poor feeding affects fish health and degrades water quality
- No way to monitor or feed remotely when the owner is away
- A reliable automated system is needed to solve this

💡 **Solution: An IoT-based Automated Fish Feeder using ESP32**

Motivation & Objectives

MOTIVATION

- Reduce manual effort in everyday pet care
- Use IoT technology for smarter home automation
- Ensure fish are fed on time, even when away

OBJECTIVES

- Build a low-cost, reliable system using ESP32
- Support both manual and scheduled feeding
- Enable remote control via WiFi web interface
- Show real-time status on an LCD display
- Accurately track time using an RTC module

Proposed Solution

An IoT-based Automated Fish Feeder — controlled via ESP32 with WiFi connectivity.

ESP32 as the Main Controller

Handles WiFi, scheduling, servo control and LCD output in one chip.

Web-based Dashboard

A browser interface lets users trigger feeding manually or set schedules.

Scheduled & Manual Feeding

RTC module provides accurate time; feeding runs on schedule or on demand.

LCD Status Display

16×2 LCD shows live time, date and last feeding status at a glance.

System Architecture – Block Diagram

Hardware & Software Description

HARDWARE

ESP32
Main controller with built-in WiFi

RTC (DS3231)
Accurate real-time clock module

Servo Motor
Rotates to dispense food portion

LCD 16×2
Displays time, date and status

SOFTWARE

Arduino IDE (C++)
Development environment & compiler

WiFi & WebServer
Handles ESP32 network & HTTP server

RTClib
Reads and manages RTC time data

LCD & Servo Libraries
Controls display and motor output

Working Principle

01

Connect to WiFi

ESP32 boots and joins the local WiFi network

02

Fetch Time from RTC

DS3231 provides accurate current time & date

03

Update LCD

Screen shows live time, date and system state

04

Check for Trigger

Schedule match or manual web input fires feeding

05

Servo Dispenses Food

Motor rotates to release a measured food portion

06

Log the Event

Feeding time and mode are recorded for review

Results & Advantages

RESULTS ACHIEVED

- Accurate automatic feeding triggered at pre-set intervals
- Reliable multi-schedule configuration works consistently
- Web dashboard live with manual feed button and status display

KEY ADVANTAGES

Low Cost

Easy to Use

Remote Control

Reliable

Limitations & Applications

LIMITATIONS

- Requires an active WiFi network to function remotely

APPLICATIONS

- **Home Aquariums** — Automated feeding for personal fish tanks
- **Fish Farms** — Scheduled feeding across multiple tanks
- **Smart Pet Care Systems** — Part of a broader IoT home automation setup

Conclusion & Future Scope

CONCLUSION

- Successfully designed and implemented an automated fish feeding system
- Ensures timely, consistent feeding — reducing manual human effort
- Web dashboard enables full remote monitoring and manual control

FUTURE SCOPE

Mobile App

Native Android/iOS app for control on the go

Camera Monitoring

Live video feed of the fish tank

AI-based Feeding

Smart feeding based on fish behaviour patterns

Battery Backup

Uninterrupted operation during power cuts

Cloud Storage

Store and analyse feeding history remotely

Thank You

Questions & Discussion

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